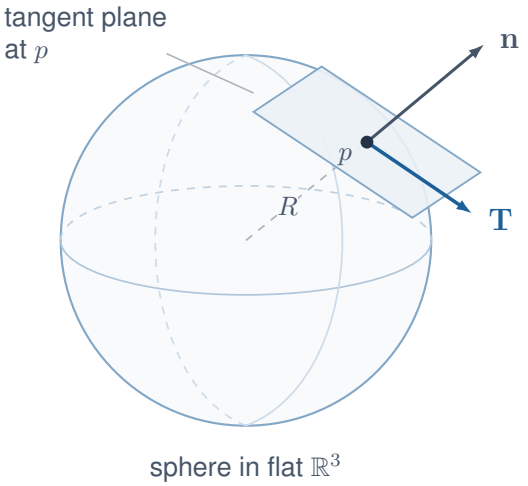
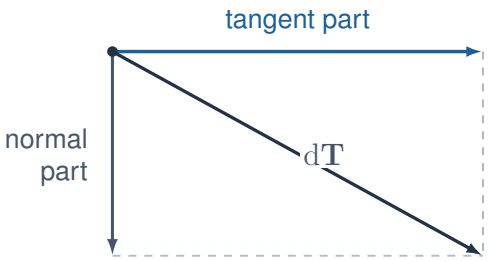
A sphere: one derivative, two geometric parts

Classical teaching derivation. An ambient derivative can leave the tangent plane.



sphere in flat $\mathbb{R}^3$

Split the change of a tangent arrow



$$\mathrm{d}\mathbf{T} = (\mathrm{d}\mathbf{T})^{\top} + (\mathrm{d}\mathbf{T})^{\perp}$$

Coordinate tangents make the split explicit

Set $\mathbf{T}_a = \partial_a \mathbf{x}$. Their ambient derivatives have a tangent part and a normal part:

$$\partial_a \mathbf{T}_b = \underbrace{\Gamma^c{}_{ab}\mathbf{T}_c}_{\text{tangent connection}} + \underbrace{\mathrm{II}_{ab}}_{\text{normal bending}} .$$

II_{ab} is a **normal vector**. A scalar second fundamental form requires a chosen normal.

Compute it for the round sphere

$$\mathbf{x} = R(\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta), \qquad \mathbf{n} = \mathbf{x}/R.$$

$$g_{ab} = \langle \mathbf{T}_a, \mathbf{T}_b \rangle = \mathrm{diag}(R^2,\ R^2\sin^2\theta), \qquad 0 < \theta < \pi.$$

$$\partial_\theta \mathbf{T}_\theta = -R\mathbf{n},$$

$$\partial_\phi \mathbf{T}_\phi = -\sin\theta\cos\theta\,\mathbf{T}_\theta - R\sin^2\theta\,\mathbf{n},$$

$$\partial_\theta \mathbf{T}_\phi = \cot\theta\,\mathbf{T}_\phi \quad (= \partial_\phi \mathbf{T}_\theta).$$

$$\mathrm{II}_{\theta\theta} = -R\mathbf{n}, \qquad \mathrm{II}_{\phi\phi} = -R\sin^2\theta\,\mathbf{n}, \qquad \mathrm{II}_{\theta\phi} = 0.$$

$$K = K_{\text{ambient}} + \frac{\langle \mathrm{II}_{\theta\theta}, \mathrm{II}_{\phi\phi}\rangle - \|\mathrm{II}_{\theta\phi}\|^2}{\det g} = 0 + \frac{1}{R^2}.$$

$R = 1 \Rightarrow K = 1; \qquad R = \frac{1}{2} \Rightarrow K = 4$ (standard quantum metric).

Surface	$\lvert\kappa_1\rvert$	$\lvert\kappa_2\rvert$	K
Plane	0	0	0
Cylinder	$1/R$	0	0
Sphere	$1/R$	$1/R$	$1/R^2$

Signed curvatures flip with the normal; $K = \kappa_1\kappa_2$.

Connections act on different objects

Christoffel: tangent arrows. Berry: state fibers. A matrix form alone does not imply non-Abelian Berry physics.

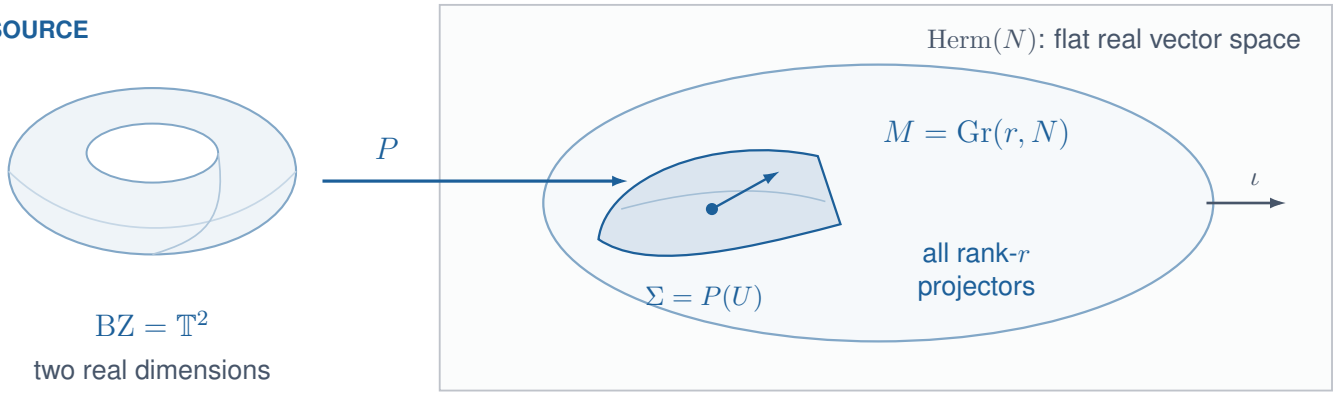
Gauss equation and second fundamental form: Avdoshkin–Popov, arXiv:2205.15353, Appendix D. The sphere calculation is a classical teaching derivation.

Blue: reference geometry. Slate: worked derivations. All diagrams are original schematics.

Which ambient space? Two stages of the same map

The intrinsic metric stays fixed; its Gauss decomposition depends on the surrounding space.

SOURCE



$U \subset \mathrm{BZ}$ is a regular patch and $\Sigma = P(U)$. Global self-intersections are allowed; the full BZ map need not be an embedding.

Two bands: one metric, two Gauss decompositions

$$P = \tfrac{I+\mathbf{n}\cdot\boldsymbol{\sigma}}{2}, \quad |\mathbf{n}| = 1; \qquad M = \mathbb{C}P^1 \simeq S^2 \quad (r = 1,\ N = 2).$$

$$G(X, Y) = \tfrac{1}{2}\,\mathrm{Tr}(XY) \quad \Longrightarrow \quad \mathrm{d}s^2 = \tfrac{1}{4}\,\mathrm{d}\mathbf{n}\cdot\mathrm{d}\mathbf{n}, \quad R_{\mathrm{eff}} = \tfrac{1}{2}.$$

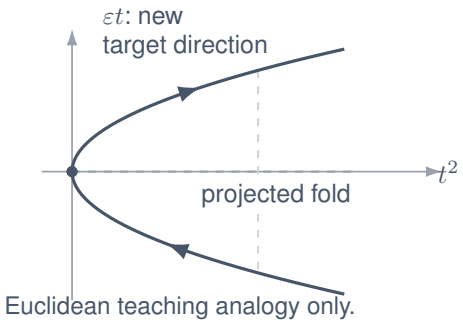
Ambient for the same regular patch	K_{ambient}	K_{II}	K_G
Inside $\mathbb{C}P^1$	4	0	4
Inside flat $\mathrm{Herm}(2)$	0	4	4

A regular two-band patch fills $\mathbb{C}P^1$ locally: $\mathrm{II} = 0$ there, despite having one unoccupied band.

A third band adds room inside state space

For $N = 3$ total bands and occupied rank $r = 1$, $M = \mathbb{C}P^2$ has four real dimensions. The regular image has two; new normals in M are possible. Berry remains $U(1)$.

Extra target directions can lift a projected fold



$$f(s, t) = (s, t^2), \quad g = \mathrm{diag}(1, 4t^2),$$

$\det g = 4t^2$ (zero at the fold).

$$f_\varepsilon(s, t) = (s, t^2, \varepsilon t), \quad \varepsilon \neq 0,$$

$$g_\varepsilon = \mathrm{diag}(1, 4t^2 + \varepsilon^2),$$

$\det g_\varepsilon > 0$: the differential regains rank.

Extension preview: θ_K is the signed Kähler angle; regular points only.

$$\Omega_{xy} = 2\cos\theta_K\sqrt{\det g}, \qquad K_G = K_{\mathrm{Gr}} + K_{\mathrm{II}}.$$

[H2] Huang, arXiv:2608.06777, §§II, IV; [H3] Huang, arXiv:2510.15760, §§II–V.

[MO] Mera–Ozawa, arXiv:2103.11583, §II, Appendix A. K_{Gr}: target sectional curvature on the image tangent plane.

Blue: known material [H2]/[H3]/[MO]. Slate: teaching derivation. Green: proposed interpretation, no novelty claim.